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Intercultural Competence and Multicultural Teaching in AI-Integrated University Classrooms: A Systematic Review of Evidence and Implications for ASEAN

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Abstract

This systematic review synthesizes 45 empirical studies (2021–2025) to examine how faculty intercultural competence mediates algorithmic bias in AI-integrated higher education across ASEAN. Current literature reveals a disconnect: research on digital intercultural competence largely ignores the algorithmic nature of educational platforms, just as studies on algorithmic bias overlook the intercultural factors that might mitigate such biases. Following PRISMA guidelines, thematic analysis of the corpus identifies three operationalized competence dimensions: technical and sociolinguistic literacy, online pedagogical adaptation, and critical algorithmic awareness, and maps them against a three-level regional digital divide spanning hardware infrastructure, ICT literacy, and sovereign computational capacity. Two propositions receive cross-methodological support: un-audited AI tools are associated with systematic penalization of non-native linguistic variation, and fragmented digitization policies are associated with constraints on regional harmonization of inclusive instructional standards. The findings indicate a need for mandatory algorithmic auditing, integrated faculty development curricula, and multilateral policy coordination through SEAMEO and the ASEAN University Network.

Introduction

Higher education institutions across Southeast Asia face a structural tension that has sharpened considerably since 2020: regional digitization mandates, formalized under the ASEAN Community Vision 2025, are accelerating the adoption of AI-assisted instructional systems at a pace that outstrips the development of culturally responsive pedagogical frameworks. The COVID-19 pandemic accelerated a structural transition that exposed pre-existing infrastructural disparities within and across member states. Institutions in metropolitan centers absorbed complex learning management systems with relative ease; universities in rural or economically peripheral territories encountered interconnected obstacles, unstable power supply, hardware shortages, and faculty without adequate digital preparation. These inequities preceded the pandemic; the crisis made them institutionally impossible to defer.

Integrating technologies such as automated essay scoring (AES), predictive analytics, and generative AI has introduced a new class of risks to these institutions. These systems are calibrated on training datasets that predominantly reflect Western-centric academic norms, linguistic conventions, and cultural referents (Jin et al., 2025). In ASEAN classrooms characterized by rich sociolinguistic diversity, this calibration is associated with systematic exclusionary outcomes: performance penalties for regional English variations, marginalization of indigenous knowledge frameworks, and replication of historical socioeconomic inequities through predictive tracking. A particularly acute manifestation is documented by Liang et al. (2023): widely deployed GPT detection tools misclassify over 50% of non-native English writing samples as AI-generated while maintaining near-perfect accuracy for native English texts. For ASEAN students whose academic writing reflects regional linguistic conventions, this bias carries direct consequences for institutional integrity.

The GPT detection bias documented by Liang et al. (2023) warrants close attention given its differential impact on ASEAN student populations. While AES bias affects grade outcomes, GPT detection bias is associated with false allegations of academic misconduct, with consequences that include disciplinary proceedings, grade reductions, and permanent annotations on academic records. When a tool associated with false positives for non-native English writers at rates exceeding 50% is embedded in institutional academic integrity policy, the practical effect is a systematic presumption of dishonesty directed at the student population that is most linguistically diverse and most educationally vulnerable. The absence of explicit institutional policies prohibiting sole reliance on these tools in ASEAN universities represents a governance gap with direct equity implications for affected student cohorts.

Consequently, faculty face dual pressures: institutional mandates for rapid technological adoption and the pedagogical necessity of cultural responsiveness. The current literature, however, treats these demands as separate concerns: studies of intercultural competence in digital environments rarely address the algorithmic properties of the platforms through which that competence must be exercised, while studies of algorithmic bias in education seldom engage with the intercultural competence constructs that could theoretically mediate those biases. Both streams are bridged here by constructing an integrated analytical frame that positions faculty digital multicultural competence as the mediating variable between algorithmic inputs and regional educational equity outcomes.

This synthesis is particularly timely given the rapid diffusion of AI across regional higher education systems. Between 2022 and 2025, generative AI tools transitioned from experimental supplements to default institutional infrastructure across ASEAN, with university administrators mandating integration into course delivery and assessment at a pace that has exceeded the development of faculty preparation programs and procurement governance frameworks. Structural biases embedded in these systems operate across widely deployed platforms, with implications for the academic credentials of student cohorts that will comprise the regional workforce over the next three to four decades.

Three objectives organize the synthesis. The first maps how digital multicultural competence is conceptualized and operationalized for university faculty in the post-2021 literature. The second evaluates the pedagogical opportunities and intercultural risks associated with deploying AI tools in diverse higher education settings. The third contextualizes these associations within ASEAN's geopolitical and infrastructural realities, examining constraints on regional policy harmonization. Three research questions guide the analysis: (RQ1) How is digital multicultural competence defined and operationalized for university faculty in the 2021–2025 empirical literature? (RQ2) What pedagogical opportunities and intercultural risks are associated with AI-integrated instruction in linguistically diverse settings? (RQ3) What infrastructural and policy constraints shape regional harmonization of inclusive digital instructional standards across ASEAN?

Literature Review

Foundational Frameworks and Digital Extensions

Byram's Intercultural Communicative Competence model (Byram, 1997), as operationalized in higher-education telecollaboration contexts by Gutiérrez-Santiuste and Ritacco-Real (2023), conceptualizes intercultural capacity as a co-moving configuration of attitudes, knowledge, interpretive skills, and critical cultural awareness. Its core limitation in AI-mediated instruction is the assumption of a human interlocutor capable of contextual adjustment: automated grading engines and generative content platforms cannot negotiate sociolinguistic complexity; they replicate the normative structures of their training data. Deardorff's (2006) process model identifies openness and tolerance of ambiguity as attitudinal preconditions for cultural transformation, dispositions that, in digital contexts, must be supplemented by the capacity to interrogate the computational systems mediating instruction.

Quantitative evidence from Sultan et al. (2025), based on a survey of philology students (n = 334), finds that AI literacy and cultural literacy jointly explain 23.7% of the variance in intercultural communication proficiency, confirming these competencies are analytically distinct rather than interchangeable. Choompunuch et al. (2024) demonstrate that intercultural communicative competence must be embedded in formalized pre-service preparation rather than left to incidental professional experience. Cancino and Núñez (2023) provide further empirical grounding through qualitative analysis of EFL teachers in multicultural classrooms, confirming close associations between instructors' intercultural sensitivity and interactive performance in culturally heterogeneous settings. Samuel et al. (2023) argue, through the Culturally Adaptive Thinking in Education (CATE) framework, that AI education itself must be redesigned around culturally adaptive principles to serve students whose epistemological frameworks are not represented in the dominant training data.

Structured Digital Modalities and Intercultural Outcomes

The distinction between structured and unstructured digital contact matters for institutional investment. COIL remains the most rigorously evaluated modality for intercultural competence development in higher education: Hackett et al. (2023), in a quasi-experimental study across nine institutional partnerships, report a statistically significant pre-to-post increase in student cultural intelligence (Cohen's d = 0.61, 95% CI [0.44, 0.78]), with gains attributable to explicit task design, mandatory reflection cycles, and facilitated debriefing rather than digital co-presence. An institution that funds bandwidth without funding facilitation is purchasing preconditions for intercultural contact without the pedagogical structures that link that contact to learning. Antonenko and Abramowitz (2023) find consistent results in teacher education contexts, where programs that separate AI technical competence from equity considerations are associated with weaker faculty preparedness. Watted and Barak (2024) confirm that learner agency and social interaction embedded in course architecture exhibit associations with differential engagement outcomes, a finding with direct implications for ASEAN institutions where passive broadcast formats remain the institutional default.

Generative AI Integration and Institutional Policy

The consequences of this governance gap are not merely administrative. When faculty lack the frameworks to evaluate AI outputs for cultural assumptions, the defaults of generative platforms become the de facto curriculum. Crompton and Burke (2023), in a systematic review of AI integration across higher education institutions, find that faculty who deploy AI tools without structured critical evaluation frameworks tend to replicate, rather than interrogate, the cultural and epistemological assumptions embedded in platform defaults, a preparation gap that current professional development frameworks do not systematically address.

The policy asymmetry documented by Jin et al. (2025) has a reinforcing effect within ASEAN: institutions in member states with less developed AI governance frameworks are simultaneously the most likely to adopt commercial AI tools without cultural adaptation and the least likely to have faculty preparation programs that develop the critical algorithmic awareness needed to compensate. Yusuf et al. (2024), mapping 407 publications on generative AI in education across Web of Science, Scopus, and ScienceDirect, identify a persistent gap in research on the cultural dimensions of GenAI, confirming that equity and cultural specificity remain among the least-addressed concerns in the global literature. Zawacki-Richter et al. (2019), in a systematic review of AI applications in higher education, find that the educator perspective remains systematically underrepresented in the evidence base relative to technical and student-facing studies. O'Dowd and Dooly (2022), examining teacher professional development through virtual exchange, caution that the rapid institutionalization of digital international formats risks privileging access metrics over pedagogical depth, a dynamic now replicated at a larger scale in AI integration policy. This pattern is associated with a risk concentration where the institutions serving the most linguistically and culturally diverse student populations operate with the weakest protective frameworks against the algorithmic biases that most directly threaten those populations.

Culturally Responsive Pedagogy in Online Environments

Rozi et al. (2025) confirm, through a systematic review, that culturally congruent instructional strategies are positively associated with students' critical-thinking performance across institutional and national contexts. Shadiev et al. (2024), in a systematic review of 31 review studies spanning 2003–2023, confirm that technology-supported cross-cultural learning is most effective when explicit cultural comparison tasks and structured reflection protocols are embedded in the design. These findings converge with the COIL evidence base and directly challenge the assumption that digital co-presence alone is sufficient for intercultural development. Vincent-Layton (2022) establishes that digital Culturally Responsive Teaching (CRT) requires deliberate, non-neutral course design: proactive selection of pluralistic curricular resources, structural neutralization of technological access barriers, and active facilitation of equitable participation in asynchronous discussions. Johnson (2022) argues that structural embedding of equity metrics in course design, rather than additive diversity statements in syllabi, yields the more consequential intervention. Dzerviniks et al. (2024), comparing Online Multicultural Teaching Competency across twelve institutional contexts, find that instructor readiness is

highly variable and systematically dependent on institutional scaffolding rather than individual faculty disposition, a finding that implicates professional development policy as the primary leverage point.

The structural bifurcation between AI technical training and equity preparation is documented by multiple studies. Dolah et al. (2025) confirm that AI integration within learning management systems correlates with measurable instructional friction when faculty lack the capacity to moderate automated features, and that affected instructors default to passive lecture delivery, disabling the interactive structures through which culturally responsive online pedagogy functions.

Algorithmic Bias: Mechanisms and Evidence

Baker and Hawn (2022) establish the foundational empirical record: predictive models and automated assessment systems exhibit performance disparities that correspond to historical societal inequities, and these disparities are structural features of systems calibrated on non-representative training data rather than incidental artifacts. Litman et al. (2021) demonstrate that natural language processing parsers in AES systems systematically underrate non-standard syntactical structures and regional narrative variations even when content quality is held constant, confirming that the performance penalty is linguistic and cultural rather than intellectual. Gándara et al. (2024) demonstrate, through analysis of nationally representative US college data, that predictive models consistently yield less accurate results for racially minoritized students and that common bias-mitigation techniques are generally ineffective at eliminating these disparities across racialized groups, meaning bias is structurally embedded rather than correctable through aggregate adjustments.

Liang et al. (2023) extend this analysis to GPT detection tools, documenting misclassification rates exceeding 50% for TOEFL essays from non-native English writers, while approaching zero for native English samples. The mechanism is a training data asymmetry: detectors learn to associate the constrained lexical variety and syntactic patterns characteristic of non-native academic writing with AI generation. In ASEAN contexts where English functions as a second or third language for the majority of students, this bias carries direct institutional consequences at the intersection of academic integrity policy and assessment equity. In admissions and resource allocation, Gándara et al. (2024) show that black-box predictive analytics encode structural exclusion by yielding systematically less accurate predictions for racially minoritized student populations, and that bias-mitigation techniques show limited effectiveness across multiple fairness criteria. Bird et al. (2025) confirm that predictive models correlate algorithmic underestimation of success with Black student subgroups and that the magnitude of this bias varies substantially depending on how institutional risk thresholds are defined.

Generative content tools introduce a related but distinct risk. When deployed without critical guidance, these platforms default to Eurocentric cultural referents and Western disciplinary conventions, marginalizing indigenous and regional knowledge systems (DaCosta & Kinsell, 2024). Lu et al. (2024) find, through a mixed-methods analysis, that ChatGPT-assisted grading of undergraduate academic writing exhibits systematic variation across demographic subgroups, with students from non-dominant

linguistic backgrounds reporting lower satisfaction with the quality of machine-mediated feedback. Thelwall (2024) confirms, through a systematic review, that automated research assessment tools exhibit design biases that co-vary with institutional prestige variables, effectively encoding existing academic hierarchies into algorithmic form.

Research Gap

These three literature streams each address part of the problem, but none address the intersection. Intercultural competence research does not systematically engage with the algorithmic properties of the platforms it studies; algorithmic bias research does not draw on intercultural constructs that might moderate those biases; and ASEAN educational technology scholarship has largely treated AI as a value-neutral administrative instrument rather than a sociolinguistically loaded one. Institutions making AI procurement decisions draw on educational technology literature that treats AI as neutral, while institutions designing faculty development draw on intercultural competence literature that does not address algorithmic properties. Faculty are consequently unprepared on two fronts: they lack the competence to mediate algorithmic bias, and they operate within governance frameworks that do not audit AI outputs on their behalf. The integrated framework proposed here addresses both simultaneously by positioning faculty digital multicultural competence as the mediating variable between algorithmic inputs and regional equity outcomes.

Conceptual Framework

The framework advances a position on AI's institutional role that diverges from both its critics and its advocates in the current policy discourse. AI skeptics in educational contexts tend to treat algorithmic tools as categorically incompatible with culturally responsive pedagogy, a position the evidence does not support, since structured digital environments with active faculty mediation are associated with measurable intercultural gains. AI advocates tend to treat these tools as categorically equity-enhancing, a position equally unsupported by the evidence, since un-audited algorithmic deployment is consistently associated with performance penalties for linguistically marginalized student populations. The framework positions the equity direction of AI as contingent and negotiable, a function of governance frameworks, faculty preparation, and infrastructural preconditions rather than an intrinsic property of the technology.

Three structural layers organize the framework: The input layer conceptualizes AI-integrated educational systems as active, value-laden computational agents whose outputs reflect the demographic and cultural composition of their training data, rather than as neutral administrative instruments. This reconceptualization is the central theoretical contribution, supported by the evidence on convergent bias reviewed above. At the mediation layer, faculty digital multicultural competence operates through three analytically distinct dimensions. Technical and sociolinguistic literacy encompasses the use of digital tools to proactively establish linguistic safety: normalizing accent variation, accommodating multiple rhetorical traditions in discussion design, and managing dominant-voice dynamics in collaborative

spaces. Online pedagogical adaptation encompasses the deliberate selection of pluralistic curricular resources, the equitable structuring of digital participation, and multiple assessment pathways that accommodate varying levels of technological access. Cots et al. (2022), examining language awareness and intercultural competence in multilingual educational contexts, confirm that explicit attention to linguistic diversity in course design is positively associated with students' intercultural sensitivity outcomes across institutional settings. Critical algorithmic awareness encompasses the capacity to audit AI outputs for cultural stereotypes, design inclusive prompts that mitigate training-data biases, conduct disaggregated assessment reviews, and evaluate GPT detection tools for false-positive rates among non-native-speaker populations. Alam (2022) documents that adaptive learning systems in higher education contexts exhibit their most equitable outcomes when human pedagogical oversight is explicitly built into deployment frameworks rather than treated as optional monitoring. Zambrano et al. (2021), examining the interaction between prior knowledge and information presentation in collaborative learning tasks, demonstrate that heterogeneous student groups exhibit greater performance variation under standardized digital assessment than under human-evaluated tasks, a finding with direct implications for AI deployment in culturally diverse cohorts.

At the output layer, the interaction between algorithmic inputs and faculty mediation is associated with two contrasting outcome trajectories: equitable human capital development, in which culturally inclusive deployment and competent mediation are present; and workforce marginalization, in which un-audited standardization and insufficient mediating capacity predominate. The ASEAN digital divide serves as a structural moderator that conditions the extent to which institutional interventions can achieve equitable outcomes. Two propositions anchor the evaluative analysis: Proposition 1 (P1) states that un-audited generative AI and automated assessment tools are associated with the systematic penalization of non-native linguistic variation; Proposition 2 (P2) states that fragmented national digitization policies constrain regional harmonization of inclusive instructional standards.

Figure 1 presents the three-layer conceptual framework that this review uses to organize the relationship among AI systems, faculty competence, and regional equity outcomes across ASEAN.

Research Methodology

Search Strategy

This study employs a systematic integrative review design (Torraco, 2016) and follows PRISMA 2020 reporting guidelines to ensure transparency in search and screening procedures. Scopus and Web of Science were searched using the string: ("intercultural competence" OR "cultural intelligence" OR "multicultural teaching" OR "culturally responsive pedagogy") AND ("higher education" OR "university faculty" OR "professors") AND ("artificial intelligence" OR "algorithmic" OR "digital learning" OR "online learning" OR "virtual exchange" OR "COIL"). The search was restricted to peer-reviewed English-language journal articles published between January 2021 and December 2025.

Inclusion and Exclusion Criteria

Articles were included if they met four criteria: empirical design (qualitative, quantitative, or mixed-methods) published in a peer-reviewed journal; higher education focus examining university faculty, instructional practices, or course design; engagement with digital or AI-integrated learning environments; and examination of intercultural competence, culturally responsive pedagogy, or multicultural teaching as a core conceptual focus. Articles were excluded if they addressed K–12 contexts; represented purely technical computer science work without pedagogical evaluation; or were classified as editorials, book reviews, or non-peer-reviewed abstracts. All criteria were established prior to the search to prevent post-hoc inclusion bias.

Screening and Data Extraction

The initial search returned 1,248 records. Following duplicate removal, 782 records underwent title and abstract screening by two independent raters (Cohen's Kappa = 0.86 for the screening phase); 612 were excluded, yielding 170 full-text articles for eligibility assessment. A further 125 were excluded: 54 for insufficient focus on higher education faculty, 41 for lacking an empirical focus on intercultural or multicultural teaching, and 30 for purely technical algorithmic orientations without educational analysis. The final synthesis corpus comprised N = 45 empirical articles. A structured data extraction matrix captured, for each article, the author, year, geographic locus, research design, sample characteristics, digital or AI context investigated, and core intercultural and pedagogical findings.

Of the 45 included articles, 18% are situated within ASEAN contexts and 35% in the United States and Europe, a bibliometric asymmetry that reflects structural constraints on regional research capacity rather than differential relevance. This geographic imbalance is itself evidence for P2 and requires that recommendations be applied to ASEAN contexts with attention to their derivation from higher-resource settings.

Thematic Synthesis

Data management across the 45 included studies used a purpose-built extraction matrix capturing: author(s) and year, geographic locus, research design, sample characteristics, AI context, and core intercultural findings. Coding was conducted manually by two independent raters; discrepancies were resolved through structured deliberation against pre-specified inclusion criteria. Initial screening inter-rater reliability reached $\kappa = 0.86$, indicating substantial agreement (Landis & Koch, 1977); disagreements over full-text eligibility (n = 12) were resolved through arbitration prior to thematic synthesis. Thematic synthesis proceeded through three iterative coding cycles. Descriptive codes were first assigned to all passages addressing digital multicultural competence, AI capabilities and risks, and regional infrastructural constraints. These were then organized into conceptual categories aligned with RQ1 (faculty competence operationalization), RQ2 (pedagogical opportunities and intercultural risks), and RQ3 (regional digital divide constraints). In the third cycle, inter-category relationships were mapped against Propositions 1 and 2 to evaluate direction, consistency, and methodological breadth across qualitative, quantitative, and mixed-methods designs. Meta-analytic effect size pooling was not attempted due to structural heterogeneity across ASEAN national policy contexts. The approach is a

theoretically grounded narrative synthesis with systematic search procedures and structured coding protocols.

Figure 2 shows the PRISMA 2020 flow diagram detailing the search, screening, and inclusion process through which 45 studies were selected for synthesis.

Results

Descriptive Characteristics

Publication volume in the 45 included studies increases sharply between 2023 and 2025, co-varying with the post-pandemic consolidation of hybrid learning and the commercial diffusion of large language model tools. Qualitative designs account for 40% of the corpus, mixed-methods for 35%, and quantitative studies for 25%. Geographically, the literature from 2021 onward shows a shift toward China, Israel, Australia, and ASEAN member states, the latter representing 18% of the corpus. Subject domain concentration is in language learning (35%), social sciences (30%), and teacher education (20%), with a smaller but expanding representation in STEM.

Table 1 provides a descriptive and methodological synthesis of representative studies from the corpus, including primary method, geographic context, and core finding relevant to the review propositions.

Table 1
Descriptive and Methodological Synthesis of Representative Reviewed Literature

Author / Year	Method	Context	Core Finding
Hackett et al. (2023)	Quasi-Experimental	COIL	Cultural intelligence increased significantly (d = 0.61); structured facilitation, not digital co-presence, is the proximate variable.
Sultan et al. (2025)	Quantitative Survey	AI and Cultural Literacy	AI literacy and cultural literacy jointly explain 23.7% of the variance in intercultural communication proficiency.
Litman et al. (2021)	Quantitative	Automated Essay Scoring	NLP parsers systematically underrate non-standard syntax and regional narrative variation independent of content quality.
Liang et al. (2023)	Quantitative	GPT Detection Bias	Seven GPT detectors misclassify > 50% of non-native-English TOEFL essays as AI-generated; achieve near-perfect accuracy on native samples.
Jin et al. (2025)	Policy Analysis	GenAI Institutional Adoption	AI policy is concentrated in the Global North; Southeast Asian institutions exhibit markedly lower levels of governance engagement.
Dzerviniks et al. (2024)	Comparative	Online Multicultural Teaching	Instructor readiness is highly variable and dependent on institutional scaffolding rather than individual disposition.
Dolah et al. (2025)	Systematic Review	AI in LMS	AI-LMS integration correlates with instructional friction when faculty lack the capacity to moderate automated features.
Baker & Hawn (2022)	Systematic Review	Algorithmic Bias	Predictive models exhibit performance disparities that align with historical societal inequities; these disparities are structural, not incidental.
Lu et al. (2024)	Mixed-Methods	AI Grading Fairness	Machine-mediated grading correlates with a sharp decline in perceived fairness among students from non-dominant linguistic backgrounds.

Note: IC = Intercultural Competence; MCT = Multicultural Teaching; LMS = Learning Management System; AES = Automated Essay Scoring. A thematically selected cross-section of the 45 synthesized studies.

The geographic distribution of the 45 included studies reveals a pattern that itself indicates the research gap this review addresses. ASEAN-situated empirical research accounts for 18% of the corpus, while United States and European studies account for 57%, a disparity that reflects not the relative importance of the research question across regions but the differential research capacity associated with the first and third levels of the digital divide. The institutions with the greatest capacity to generate peer-reviewed English-language research indexed in Scopus and Web of Science are concentrated in the same high-

resource institutional environments that face attenuated versions of the equity challenges this review examines. The policy and practice recommendations emerging from this literature base must therefore be applied to ASEAN contexts with explicit attention to their derivation from research settings with substantially different resource endowments, governance capacities, and student population characteristics.

RQ1: Operationalizing Digital Multicultural Competence

Thematic synthesis identifies three empirically co-dependent dimensions. Technical and sociolinguistic literacy encompasses the use of digital tools to establish linguistic safety, including normalizing accent variation, structuring discussion prompts to accommodate multiple rhetorical traditions, and managing dominant-voice dynamics in collaborative spaces. Deficits in this dimension are associated with higher rates of stereotype activation and misinterpretation in asynchronous text-based formats. Online pedagogical adaptation operationalizes culturally responsive teaching principles within LMS constraints: the deliberate selection of diverse curricular resources, the equitable structuring of digital participation, and flexible assessment pathways that accommodate varied access conditions. Dzerviniks et al. (2024) confirm that this dimension is the most variable across contexts and most strongly associated with investment in institutional support. Critical algorithmic awareness encompasses the capacity to audit AI outputs for hidden cultural stereotypes, construct inclusive prompts that attenuate training data biases, conduct disaggregated assessment reviews, and evaluate GPT detection tools for non-native-speaker false-positive rates (Liang et al., 2023). Liu (2024) confirms that ethical and human-centered AI competence components receive the least attention in existing teacher preparation curricula across all five dimensions of teacher AI literacy. Janssen et al. (2022) confirm that technical operational competence and critical evaluative competence are analytically distinct and independently require investment, while Manca and Ranieri (2022) find that faculty AI integration challenges co-vary with institutional support structures rather than individual attitudes. Parrish and Linder-VanBerschot (2022) identify instructional design as the primary mediator through which cultural assumptions are operationalized or challenged in distributed online learning.

Across the three dimensions, competence development is primarily a structural rather than a dispositional outcome. Dzerviniks et al. (2024) find that instructors at institutions with structured multicultural teaching support demonstrate substantially stronger competence profiles than those at comparable institutions without structured support. The difference reflects professional learning ecosystems rather than individual aptitude. Institutional accountability frameworks, not individual faculty development mandates, constitute the appropriate governance response.

RQ2: Pedagogical Opportunities and Intercultural Risks

The corpus reveals an asymmetric pattern. Adaptive learning systems offer personalized pathway development, particularly valuable in large ASEAN cohorts where differentiated instruction has historically been resource-constrained. Natural language processing applications can reduce language barriers through real-time translation and multilingual feedback. Administrative scaffolding functions,

when purposively configured, free faculty time for mentoring and cross-cultural facilitation. However, the realization of these benefits is contingent on a level of critical pedagogical oversight that current faculty preparation frameworks do not reliably develop.

The documented risks are more deeply entrenched. AES platforms impose systematic penalties on non-native English syntax and culturally distinct narrative structures, operating below the threshold of administrative visibility (Litman et al., 2021; Gándara et al., 2024). GPT detection tools misclassify non-native English writing as AI-generated at rates exceeding 50%, resulting in institutional integrity consequences that are disproportionately borne by ASEAN students (Liang et al., 2023). Predictive analytics encode structural exclusion through geographic and demographic proxies (Gándara et al., 2024; Bird et al., 2025). Generative content tools default to Eurocentric referents when not actively counteracted (DaCosta & Kinsell, 2024). These risks span multiple stages of the student experience, from admissions and ongoing assessment through to resource allocation, with the potential for compounding effects across institutional encounters that no single longitudinal study in the corpus has yet directly measured. The evidence provides consistent cross-methodological support for P1.

The conditions under which AI pedagogical opportunities are realized versus foreclosed are captured in a faculty mediation continuum. At the lower end of the continuum, where faculty possess basic technical familiarity but lack critical algorithmic awareness, AI tools tend to standardize rather than personalize and to penalize rather than accommodate linguistic variation. At the upper end, where faculty actively configure platform settings, design culturally grounded prompts, and implement disaggregated assessment auditing, the same tools exhibit associations with more equitable practice. The position of an institution on this continuum is not fixed; it is shaped by the quality of professional development, workload allocation, and the presence or absence of institutional algorithmic auditing requirements. The evidence suggests that investments in faculty preparation are associated with meaningful shifts along this continuum, altering the equity direction of AI deployment without requiring changes to the platforms themselves.

The evidence base for RQ2 converges on a finding with direct governance implications: the intercultural risks of AI deployment are structural to systems calibrated on non-representative training data, not incidental to particular platforms. Rozi et al. (2025) confirm that the cultural alignment of instructional resources is positively associated with students' critical-thinking performance across institutional contexts. When AI content generation defaults to Eurocentric referents, these performance benefits are constrained for students whose linguistic and cultural backgrounds are absent from platform training data. In ASEAN classrooms, where most students study in their second or third language, this is the structural condition rather than the exception.

Table 2 maps four AI application types against their documented intercultural risks and the evidence-based mitigation strategies identified across the corpus.

Table 2
AI Capabilities, Documented Intercultural Risks, and Evidence-Based Mitigation Strategies

AI Capability	Documented Intercultural Risk	Evidence-Based Mitigation
Automated Essay Scoring	Systematic penalty for non-native English syntax and regional narrative structures; performance deficit invisible in aggregate reporting (Litman et al., 2021; Gándara et al., 2024).	Human-in-the-loop validation stratified by students' first language; multiple-assessment-pathway design; regular disaggregated performance audits.
GPT Writing Detection	Misclassification rate > 50% for non-native English writing as AI-generated; near-zero for native English (Liang et al., 2023).	Prohibit as the sole integrity mechanism; require human reviewer confirmation for all flagged non-native-speaker submissions.
Predictive Student Analytics	Yields systematically less accurate predictions for racially minoritized student populations; common bias-mitigation techniques show limited effectiveness across fairness criteria (Gándara et al., 2024; Bird et al., 2025).	Combine predictive data with holistic qualitative review; mandate performance disaggregation by nationality and socioeconomic status in institutional reporting.
Generative Content Tools	Defaults to Eurocentric cultural referents; marginalizes indigenous and regional knowledge frameworks (DaCosta & Kinsell, 2024).	Culturally grounded prompt engineering in faculty guidelines; student assignments requiring critical analysis of AI-generated cultural representations.

Note: Risk characterizations are derived from the synthesis corpus. Mitigation strategies reflect evidence-based recommendations across included studies.

Discussion

RQ3: The ASEAN Digital Divide as Structural Moderator

Three discrete and mutually reinforcing levels of digital divide constrain inclusive multicultural instruction across ASEAN. Hardware infrastructure and connectivity constitute the first level. Singapore and select Malaysian metropolitan clusters have near-universal high-speed network penetration; institutions in Cambodia, Lao PDR, Myanmar, and peripheral zones of Indonesia and the Philippines face persistent hardware shortages, unreliable power, and limited bandwidth (Rediyasa et al., 2025; Apriliyanti et al., 2021). When administrative mandates enforce automated instructional transitions under these conditions, the consequence is institutionalized exclusion of the least-resourced student populations from the educational modalities that policy formally promotes.

Beyond infrastructural access, the distribution of ICT literacy across faculty and students presents a distinct constraint. Keiller et al. (2023), examining the design of online faculty development programs across nine sub-Saharan African countries, document barriers qualitatively different from those in high-income institutional contexts; contextual factors including infrastructure unreliability, institutional culture, and workload constraints require explicit accommodation in program design frameworks, a

finding with direct parallels in ASEAN resource-constrained environments. Students without secondary-level digital literacy preparation enter university systems that presuppose those skills, while faculty without adequate ICT training default to passive delivery, disabling the interactive mechanisms that culturally responsive online pedagogy requires (Jamil, 2024). This bifurcation is associated with a self-reinforcing cycle in which technology adoption mandates are linked to pedagogical regression in the institutions that most need support.

Sovereign computational capacity constitutes the third and least-addressed level in existing policy discourse. Engineering and maintaining high-performing large language models requires computational investment structurally inaccessible to lower-income ASEAN institutions, sustaining a dependency on externally calibrated commercial AI systems whose parameters reflect priorities set outside the region, with no institutional accountability to regional educational equity goals. Luckin et al. (2022) argue that realizing the potential of AI in education requires institutional capacity to interrogate, select, and contextually adapt AI tools, a capacity that presupposes both human expertise and organizational infrastructure unevenly distributed across ASEAN. Jin et al. (2025) confirm that Southeast Asian universities are substantially under-represented in global AI governance policy development, meaning the algorithms governing student evaluation in the region reflect priorities established primarily in the United States, Europe, and East Asia.

Table 3 summarises the three-level ASEAN digital divide, its institutional characteristics, representative country contexts, and the targeted governance responses this review recommends for each level.

Table 3
Three-Level ASEAN Digital Divide: Characteristics, Representative Contexts, and Targeted Interventions

Level	Primary Barrier	Representative ASEAN Context	Targeted Intervention
Level 1: Infrastructure	Hardware deficit, unstable power, minimal bandwidth	Cambodia, Lao PDR, Myanmar; peripheral zones of Indonesia and the Philippines	Targeted public funding for rural broadband and institutional hardware; offline-capable LMS alternatives
Level 2: ICT Literacy	Insufficient digital pedagogical fluency among faculty and students	The majority of lower-income institutions across all ten member states	Integrated faculty development merging AI technical training with culturally responsive pedagogy; digital literacy in secondary curriculum
Level 3: Sovereign Capacity	Structural dependency on externally calibrated AI systems	All member states, most acute in lower-income contexts	ASEAN-coordinated investment in regionally calibrated AI tools; multilateral Southeast Asian language corpus agreements; cultural bias auditing in procurement contracts

Note: Characterizations are derived from the synthesis corpus. Interventions reflect cross-study evidence-based recommendations.

Macroeconomic Implications and Proposition Evaluation

The evidence provides consistent support for P2. Yang and Ma (2024) confirm that sociodemographic factors exhibit strong co-variation with patterns of technological engagement in online learning across the region. The ASEAN Education Ministers' repeated high-level declarations on digital equity have not been accompanied by the operational coordination mechanisms or infrastructure subsidies required to close the first-level divide, which is the necessary precondition for progress at levels two and three. Jin et al. (2025) find that even among institutions with AI governance policies, the emphasis falls overwhelmingly on academic integrity and productivity rather than cultural equity or linguistic inclusion.

Comparative analysis of national digitization trajectories within ASEAN illustrates the mechanisms through which P2 operates at the country level. Singapore's Smart Nation initiative and Malaysia's Malaysia Digital Economy Blueprint represent the most developed frameworks for AI integration in higher education, providing institutional funding streams, regulatory guidance, and technical capacity building targeted at university administrators. These frameworks have enabled leading institutions to engage with second- and third-level digital-divide challenges, develop localized AI governance policies, and begin investing in sovereign computational infrastructure. In contrast, Cambodia, Lao PDR, and Myanmar have digitization master plans that remain at the declaratory stage without operational funding mechanisms. Institutions in these territories face first-level access constraints that render second- and third-level challenges institutionally unreachable, a qualitative difference from the capacity gaps faced by better-resourced member states.

Vietnam's National Digital Transformation Program 2025 has accelerated AI adoption at a rate that has outstripped governance frameworks; Jin et al. (2025) confirm that AI ethics and cultural equity remain absent from institutional policy documentation. Indonesia and the Philippines exhibit an intermediate pattern: leading research universities can engage with advanced governance frameworks, but archipelagic geography renders nationally developed policy structurally inaccessible to most provincial institutions.

The workforce consequences are material. Institutions with adequate infrastructure and strong faculty mediation are associated with graduates possessing the technological fluency and cultural intelligence required for transnational digital employment. Peripheral institutions, whose cohorts are assessed by systems that penalize regional linguistic variation and misclassify non-native writing as AI-generated, yield credentials that underrepresent actual intellectual capacity, intensifying intra-regional talent migration toward metropolitan hubs and contradicting the economic convergence goals of the ASEAN Community Vision 2025.

A reframing supported by the synthesis is the treatment of faculty digital multicultural competence not as a professional aspiration but as equity infrastructure, a structural condition whose absence is associated with predictable and measurable consequences for student outcomes regardless of individual faculty intentions. Institutions that deploy AI assessment systems without ensuring that faculty possess the competence to audit and effectively moderate their outputs delegate consequential academic decisions to unaudited computational agents whose cultural assumptions systematically

disadvantage students from non-dominant linguistic backgrounds. The evidence on perceived assessment quality from Lu et al. (2024) illustrates this dynamic precisely: students from non-dominant linguistic backgrounds report systematically lower satisfaction with AI-mediated grading feedback, consistent with students' awareness of differential treatment under automated assessment. The appropriate response operates at the level of assessment design, AI procurement, and faculty preparation, not communication strategies.

The evidence identifies a specific policy failure mode that ASEAN governance frameworks are currently replicating at scale. When national digitization policies mandate AI adoption without specifying cultural audit requirements, and when procurement contracts omit linguistic equity performance standards, structural biases become embedded in institutional infrastructure for the lifetime of those contracts. Each renewal cycle without equity auditing extends the period during which students from non-dominant linguistic backgrounds bear the performance penalties documented by Litman et al. (2021), Gándara et al. (2024), and Liang et al. (2023).

The convergent evidence reframes apparently isolated implementation challenges as symptoms of a single structural condition: AI systems trained on non-representative data are deployed without cultural auditing in environments lacking faculty preparation and governance infrastructure to mitigate their biases. This structural diagnosis implies a structural response. Institution-level governance frameworks and faculty preparation curricula operating across all AI-integrated instructional contexts. Member state institutions cannot individually negotiate culturally calibrated training data with global AI vendors; these are collective action problems that require coordination through SEAMEO and the ASEAN University Network, which exist precisely to facilitate such a regional response.

Limitations

Restrictions on English-language publications in Scopus and Web of Science exclude research in Bahasa Indonesia, Thai, Vietnamese, Tagalog, and other regional languages, thereby underrepresenting locally grounded ASEAN scholarship. The 2021–2025 boundary captures post-pandemic and post-ChatGPT developments but excludes pre-pandemic foundational literature. Both propositions are evaluated against synthesized secondary evidence; longitudinal, cross-institutional studies within ASEAN constitute the necessary next methodological step. The majority of empirical evidence supporting the algorithmic bias propositions derives from US, Australian, and European contexts; direct ASEAN-situated evidence on these mechanisms remains sparse, and transferability assumptions require empirical validation in regional contexts.

Cross-Regional Contextual Comparison

Situating the ASEAN findings within a broader comparative frame clarifies both the specificity of the regional challenges and the transferability of proposed responses. In the Australian and New Zealand contexts, which share structural features with advanced ASEAN economies including large international student cohorts from Southeast and East Asian countries, regulatory frameworks for AI in assessment

are more developed but similarly struggle with the linguistic equity dimensions Liang et al. (2023) document for GPT detection tools. Several Australian universities have explicitly prohibited AI writing detection tools as sole integrity mechanisms because of their documented false positive rates for non-native English writers, a policy development that ASEAN institutions have not yet widely adopted despite comparable student population characteristics.

In the European context, the EU AI Act classifies AI systems used in educational assessment as high-risk applications requiring conformity assessments, including non-discrimination testing, a regulatory precedent directly addressing the structural audit gaps Baker and Hawn (2022) and Gándara et al. (2024) identify. Implementing such a framework, however, assumes institutional technical capacity absent in most lower-income ASEAN institutions, illustrating the challenge of policy transfer without proportionate capacity-building mechanisms.

Recommendations for Research and Institutional Practice

Integrated faculty development is the most directly actionable institutional response. Replacing fragmented, short-cycle technical workshops with sustained programs that explicitly merge critical algorithmic literacy and culturally responsive pedagogy addresses the bifurcated preparation among faculty who can operate AI platforms but cannot audit their cultural assumptions. Nieto-Morales and Jimenez (2024), in a systematic review of cultural diversity training for teachers, find that sustained and structurally integrated programs yield more durable competence outcomes than short, standalone modules, with substantially larger effect sizes when training is embedded in faculty workload rather than offered as optional professional development. Adeyemi et al. (2023) confirm, through a mixed-methods analysis, that culturally responsive pedagogy is associated with improved learning outcomes across diverse higher education settings, with the magnitude of the association contingent on institutional support structures. Tlili et al. (2023), examining ChatGPT integration in educational contexts, caution that unreflective deployment is associated with the homogenization of student outputs and reduced engagement with culturally specific content; these risks are directly addressed by structured faculty mediation. Practical curriculum content should include hands-on AES performance auditing, disaggregated by students' first language; evaluation of GPT detection tools for non-native-speaker false-positive rates; and culturally grounded prompt engineering calibrated to regional linguistic communities. ASEAN pilot programs should evaluate outcomes across institutional tiers rather than solely at leading universities.

Three specific institutional mechanisms carry the strongest evidence base for mediating algorithmic bias in diverse higher education settings. Human-in-the-loop validation protocols for automated assessment, in which faculty review a stratified random sample of AI-scored submissions disaggregated by student first language and national origin, represent a minimum standard for responsible AES deployment. Stratification by linguistic background is essential: aggregate performance statistics mask the subgroup disparities that Gándara et al. (2024) demonstrate are structurally embedded in automated evaluation systems. Institutions implementing aggregate-only oversight effectively institutionalize the invisibility of these disparities, allowing structural bias to operate unchallenged by routine monitoring.

Ahsan et al. (2022), examining academic integrity in diverse higher education contexts, confirm that institutional policies relying on single automated mechanisms for detecting misconduct are associated with disproportionate adverse outcomes for non-native English-speaking students, reinforcing the case for multimodal human review.

Structured AI output auditing, the systematic comparison of AI-generated curricular content against regional knowledge frameworks and indigenous epistemological traditions, converts critical algorithmic awareness into routine pedagogical practice. When integrated into standard course design workflows rather than treated as an exceptional quality check, this practice is associated with the professional habits that Liu (2024) identifies as among the least-developed components of teacher AI literacy in existing preparation curricula. Practical implementation requires allocating dedicated time within faculty workload models and developing institution-specific auditing rubrics calibrated to the linguistic and cultural communities the institution serves. Culturally grounded prompt engineering, the deliberate specification of regional cultural contexts, indigenous knowledge frameworks, and non-Western pedagogical traditions as reference points within AI prompts, represents a third specific mechanism with demonstrated effectiveness in mitigating the Eurocentric defaults that DaCosta and Kinsell (2024) document across platform types.

Mandatory pre-deployment algorithmic auditing translates institutional accountability into operational practice. Procurement contracts should require commercial vendors to demonstrate predictive fairness and linguistic equity across the demographic subgroups represented in the purchasing institution's student population. AES platforms should undergo recalibration using regionally sourced datasets, with performance disparity reporting disaggregated by nationality, first language, and socioeconomic background as a standard accountability condition. GPT detection tools should be explicitly prohibited from serving as sole integrity mechanisms given their documented bias (Liang et al., 2023).

Multilateral policy coordination through SEAMEO and the ASEAN University Network is necessary to address what individual institutional reforms cannot reach: AI system calibration outside the region and infrastructural asymmetries that prevent regional declarations from reaching institutions most requiring support. Kim and Maloney (2021) argue that sustainable digital transformation requires alignment across governance, faculty capacity, and infrastructure simultaneously, a tripartite precondition that ASEAN institutions must address given resource constraints. Targeted investment in first-level divide reduction is a precondition, not a supplementary intervention, and regional investment in locally calibrated AI tools would begin to address sovereign capacity dependency by creating alternatives to commercially calibrated systems that underrepresent Southeast Asian linguistic patterns.

Recommendations for Future Research

Longitudinal cross-institutional empirical studies within ASEAN are the necessary next step. Mixed-methods designs combining computational NLP analysis of platform outputs with faculty and student interview data would examine specific algorithmic bias mechanisms within identified ASEAN language

communities, providing the contextually specific evidence base currently absent. Comparative studies tracking equity effects of integrated faculty development programs across institutional tiers, controlling for first-level digital divide indicators, would establish whether the interventions synthesized here are associated with measurable equity gains in diverse ASEAN contexts. Research directly testing GPT detection false-positive rates across specific Southeast Asian language communities would translate Liang et al.'s (2023) findings into regionally actionable evidence. Policy analysis of SEAMEO and AUN digitization coordination mechanisms, examining the gap between declared goals and operational implementation, would strengthen the empirical foundation for governance reform recommendations.

AI procurement cycles in ASEAN higher education typically last three to seven years. Governance frameworks adopted in the current cycle (approximately 2023–2028) will shape the equity conditions of ASEAN higher education for the remainder of the decade, establishing institutional norms resistant to revision once embedded in procurement contracts and accreditation standards. Empirical research must therefore be generated on a timeline commensurate with the pace of institutional adoption rather than conventional academic publication rhythms.

Conclusion

The ASEAN region's particular urgency derives from the intersection of three conditions rarely found together in other global higher education contexts: extreme sociolinguistic diversity within and across institutional student populations; a three-level digital divide that is associated with mutually reinforcing structural disadvantages for the majority of institutions outside metropolitan centers; and a rapid AI adoption trajectory that is outpacing governance framework development at both institutional and national levels. This intersection suggests that the equity consequences of unmediated AI deployment may be more acute in ASEAN than in the high-resource contexts where most empirical evidence has been generated, though direct comparative studies are not yet available to confirm the magnitude of this difference.

Forty-five empirical studies converge on a finding that regional educational technology policy has not yet adequately absorbed: the equity direction of AI integration in ASEAN higher education exhibits consistent positive associations with governance frameworks, cultural auditing processes applied to AI outputs, and the faculty competence available to mediate structural biases, rather than with the pace of adoption alone. Digital tools are associated with genuine intercultural benefits when deployed within structured, pedagogically purposeful frameworks mediated by faculty with well-developed digital multicultural competence. The same tools, deployed without that mediation, are associated with systematic performance penalties for non-standard linguistic variation, misclassification of non-native writing as AI-generated, Eurocentric content defaults, and socioeconomic exclusion loops in predictive analytics, risks that span multiple stages of the student experience.

The three-tier ASEAN digital divide conditions whether any institutional intervention can achieve equitable outcomes in practice. Institutional, national, and regional recommendations are most effective

when they address these structural layers sequentially rather than treating technology adoption and cultural sensitivity as independently sufficient solutions. The theoretical repositioning of AI as a value-laden sociolinguistic actor rather than a neutral medium provides the analytical foundation for future empirical work measuring the equity consequences of AI governance frameworks within specific ASEAN language communities, research that is both timely and methodologically tractable.

References

1. Baker RS, Hawn A (2022) Algorithmic bias in education. Int J Artif Intell Educ 32(4):1052–1092. https://doi.org/10.1007/s40593-021-00285-9
2. Bird KA, Castleman BL, Song Y (2025) Are algorithms biased in education? Exploring racial bias in predicting community college student success. J Policy Anal Manag 44(2):379–402. https://doi.org/10.1002/pam.22569
3. Gándara D, Anahideh H, Ison MP, Picchiarini L (2024) Inside the black box: Detecting and mitigating algorithmic bias across racialized groups in college student-success prediction. AERA Open 10:23328584241258741. https://doi.org/10.1177/23328584241258741
4. Apriliyanti DL, Alon I, Doherty B (2021) Bridging the digital divide: E-government development in Indonesia. Public Adm Dev 41(1):56–68. https://doi.org/10.1002/pad.1888
5. Choompunuch B, Charoensilp S, Pai L (2024) Fostering teacher multicultural sensitivity and intercultural communicative competence for sustainable development. J Teacher Educ Sustain 26(1):45–61. https://doi.org/10.2478/jtes-2024-0004
6. Cancino M, Nuñez I (2023) EFL teachers' perceptions towards multicultural classrooms: The role of intercultural sensitivity in developing intercultural communicative competence. rEFLections 30(3):620–646. https://doi.org/10.61508/refl.v30i3.267833
7. Crompton H, Burke D (2023) Artificial intelligence in higher education: The state of the field. Int J Educational Technol High Educ 20(1) Article 22. https://doi.org/10.1186/s41239-023-00392-8
8. Deardorff DK (2006) Identification and assessment of intercultural competence as a student outcome of internationalization. J Stud Int Educ 10(3):241–266. https://doi.org/10.1177/1028315306287002
9. Gutiérrez-Santiuste E, Ritacco-Real M (2023) Intercultural communicative competence in higher education through telecollaboration: Typology and development. Educ Inform Technol 28(11):13885–13912. https://doi.org/10.1007/s10639-023-11751-3
10. DaCosta B, Kinsell C (2024) Investigating media selection through ChatGPT: An exploratory study on generative artificial intelligence in the aid of instructional design. Open J Social Sci 12(4):22–45. https://doi.org/10.4236/jss.2024.124002
11. Dolah MR, Kharauddin MH, Amir N (2025) Systematic literature review of artificial intelligence integration in learning management systems: Lecturer readiness and acceptance. Int J Educational Technol High Educ 22(1). Article 12. https://doi.org/10.1186/s41239-025-00451-2

12. Dzerviniks J, Hashmi M, Jan S (2024) Online multicultural teaching competency (OMTC): A comparative study of educator readiness across digital environments. J New Approaches Educational Res 13(1):89–104. https://doi.org/10.7821/naer.2024.1.1512
13. Antonenko P, Abramowitz B (2023) In-service teachers' (mis)conceptions of artificial intelligence in K-12 science education. J Res Technol Educ 55(1):64–78. https://doi.org/10.1080/15391523.2022.2119450
14. Hackett S, Janssen J, Beach P, Perreault M, Beelen J, van Tartwijk J (2023) The effectiveness of Collaborative Online International Learning (COIL) on intercultural competence development in higher education. Int J Educational Technol High Educ 20(1):5. https://doi.org/10.1186/s41239-022-00373-3
15. Jamil M (2024) Digital inequalities and the cycle of educational exclusion in developing countries. Int J Educational Dev 105:102984. https://doi.org/10.1016/j.ijedudev.2023.102984
16. Jin Y, Yan L, Echeverria V, Gašević D, Martinez-Maldonado R (2025) Generative AI in higher education: A global perspective of institutional adoption policies and guidelines. Computers Education: Artif Intell., 8, Article 100348 https://doi.org/10.1016/j.caeai.2024.100348
17. Johnson A (2022) Culturally responsive teaching in higher education. Int J Equity Social Justice High Educ 1:25–29. https://doi.org/10.56816/2771-1803.1008
18. Liang W, Yuksekgonul M, Mao Y, Wu E, Zou J (2023) GPT detectors are biased against non-native English writers. Patterns 4(7) Article 100779. https://doi.org/10.1016/j.patter.2023.100779
19. Litman D, Zhang H, Correnti R, Matsumura LC, Wang E (2021) A fairness evaluation of automated methods for scoring text evidence usage in writing. In I. Roll, D. McNamara, S. Sosnovsky, R. Luckin, & V. Dimitrova (Eds.), Artificial Intelligence in Education (Lecture Notes in Computer Science, Vol. 12748, pp. 255–267). Springer. https://doi.org/10.1007/978-3-030-78292-4_21
20. Liu W (2024) Language teacher AI literacy: Insights from collaborations with ChatGPT. J China Computer-Assisted Lang Learn. https://doi.org/10.1515/jccall-2024-0030
21. Keiller L, Nyoni CN, Van Wyk C (2023) A model for design of online faculty development courses in resource-constrained environments: A heuristic evaluation approach. BMC Med Educ 23(1). https://doi.org/10.1186/s12909-023-04039-0. Article 60
22. Rediyasa IW, Amir S, Flowers Z (2025) Bridging the digital divide: A study on the challenges of e-learning implementation in rural and remote schools in Indonesia. J Social Sci Technol 3(3):130–145. https://doi.org/10.5121/jssut.2025.3308
23. Lu Q, Yao Y, Xiao L, Yuan M, Wang J, Zhu X (2024) Can ChatGPT effectively complement teacher assessment of undergraduate students' academic writing? Assessment & Evaluation in Higher Education, 49(5), 616–633. https://doi.org/10.1080/02602938.2024.2301722
24. Rozi F, Yusnadi, Suyanti RD, Ansya YA (2025) Culturally responsive teaching as a strategy to improve students' science literacy in higher education. Jurnal Penelitian Pendidikan IPA 11(6):58–67. https://doi.org/10.29303/jppipa.v11i6.11268

25. Samuel Y, Brennan-Tonetta M, Samuel J, Kashyap R, Kumar V, Kaashyap SK, Chidipothu N, Anand I, Jain P (2023) Cultivation of human centered artificial intelligence: Culturally adaptive thinking in education (CATE) for AI. Front Artif Intell 6:1198180. https://doi.org/10.3389/frai.2023.1198180
26. Sultan Y, Dautova G, Dalle J (2025) Examining the relationship among artificial intelligence literacy, cultural literacy, and intercultural communication proficiency. J Ethnic Cult Stud 12(5):345–362. https://doi.org/10.29333/ejecs/2839
27. Thelwall M (2024) Technology assisted research assessment: Algorithmic bias and transparency issues. Aslib J Inform Manage 77(1):175–194. https://doi.org/10.1108/ajim-04-2023-0125
28. Torraco RJ (2016) Writing integrative literature reviews: Using the past and present to explore the future. Hum Resour Dev Rev 15(4):404–428. https://doi.org/10.1177/1534484316671606
29. Vincent-Layton K (2022) Culturally responsive teaching practices in higher education online environments. Int J Equity Social Justice High Educ 1(1) Article 9. https://doi.org/10.56816/2771-1803.1009
30. Watted A, Barak M (2024) Motivation and engagement in MOOCs: Comparing students in xMOOC and cMOOC formats. J Educators Online 21(2). https://doi.org/10.9743/JEO.2024.21.2.4
31. Shadiev R, Sintawati W, Kerimbayev N, Altinay F (2024) Systematic review (2003–2023): Exploring technology-supported cross-cultural learning through review studies. Sustainability 16(2) Article 755. https://doi.org/10.3390/su16020755
32. Yusuf A, Pervin N, Román-González M, Md Noor N (2024) Generative AI in education and research: A systematic mapping review. Rev Educ 12(2) Article e3489. https://doi.org/10.1002/rev3.3489
33. Byram M (1997) Teaching and Assessing Intercultural Communicative Competence. https://doi.org/10.21832/9781800410251. Multilingual Matters
34. Alam A (2022) Employing adaptive learning and intelligent tutoring robots for virtual classrooms and smart campuses: Re-imagining the nature of education in the era of Industry 5.0. In: Mathur AK, Kumar V, Singh D, Jhamb A, Singh N (eds) Cognitive Behavior and Human Computer Interaction Based on Machine Learning Algorithm. Wiley, pp 395–406. https://doi.org/10.1002/9781119792581.ch20
35. Cots JM, Lasagabaster D, Garrett P (2022) Language awareness and intercultural competence: Insights from multilingual contexts. Lang Aware 31(2):113–130. https://doi.org/10.1080/09658416.2021.1903979
36. Luckin R, Holmes W, Griffiths M, Forcier LB (2022) Intelligence Unleashed: An Argument for AI in Education (2nd ed.). Pearson. https://doi.org/10.1007/978-3-030-72672-2
37. Zawacki-Richter O, Marín VI, Bond M, Gouverneur F (2019) Systematic review of research on artificial intelligence applications in higher education: Where are the educators? Int J Educational Technol High Educ 16(1) Article 39. https://doi.org/10.1186/s41239-019-0171-0
38. Landis JR, Koch GG (1977) The measurement of observer agreement for categorical data. Biometrics 33(1):159–174. https://doi.org/10.2307/2529310

39. Nieto-Morales C, Jimenez RM (2024) Cultural diversity training for teachers: A systematic literature review. Cogent Educ 11(1) Article 2361982. https://doi.org/10.1080/2331186X.2024.2361982
40. Ahsan K, Akbar S, Kam B (2022) Contract cheating in higher education: A systematic literature review and future research agenda. Assess Evaluation High Educ 47(4):523–539. https://doi.org/10.1080/02602938.2021.1931660
41. Tlili A, Shehata B, Adarkwah MA, Bozkurt A, Hickey DT, Huang R, Agyemang B (2023) What if the devil is my guardian angel: ChatGPT as a case study of using chatbots in education. Smart Learn Environ 10(1). https://doi.org/10.1186/s40561-023-00237-x. Article 15
42. Adeyemi BA, Adeyemi AO, Adeola OT (2023) Culturally responsive pedagogy and student learning outcomes in diverse higher education settings: A mixed-methods analysis. J Multicultural Educ 17(2):154–170. https://doi.org/10.1108/JME-04-2022-0042
43. Zambrano J, Kirschner F, Sweller J, Kirschner PA (2021) Effects of prior knowledge and information presentation on learning and collaborative tasks. Comput Hum Behav 114:106526. https://doi.org/10.1016/j.chb.2020.106526
44. O'Dowd R, Dooly M (2022) Telecollaboration and virtual exchange in language and intercultural education. Language Learning & Technology, 26(2), 1–10. https://doi.org/10125/73522
45. Manca S, Ranieri M (2022) Challenges and opportunities of teaching AI in higher education: Insights from faculty perspectives. Technol Knowl Learn 27(3):831–849. https://doi.org/10.1007/s10758-022-09612-6
46. Parrish M, Linder-VanBerschot J (2022) Cultural dimensions of learning: Addressing the challenges of multicultural instruction. Int Rev Res Open Distrib Learn 23(2):1–23. https://doi.org/10.19173/irrodl.v23i2.6059
47. Janssen J, Tummel C, Richert A, Isenhardt I (2022) Towards a framework for AI literacy in higher education: Competencies, challenges and future research directions. Computers Education: Artif Intell 3:100095. https://doi.org/10.1016/j.caeai.2022.100095
48. Kim J, Maloney EJ (2021) Learning Innovation and the Future of Higher Education. Johns Hopkins University. https://doi.org/10.56021/9781421440422
49. Yang X, Ma L (2024) The multi-dimensional nature of the digital divide: Socio-demographic factors and technological engagement in online learning. Comput Hum Behav 151:108012. https://doi.org/10.1016/j.chb.2023.108012

Figures

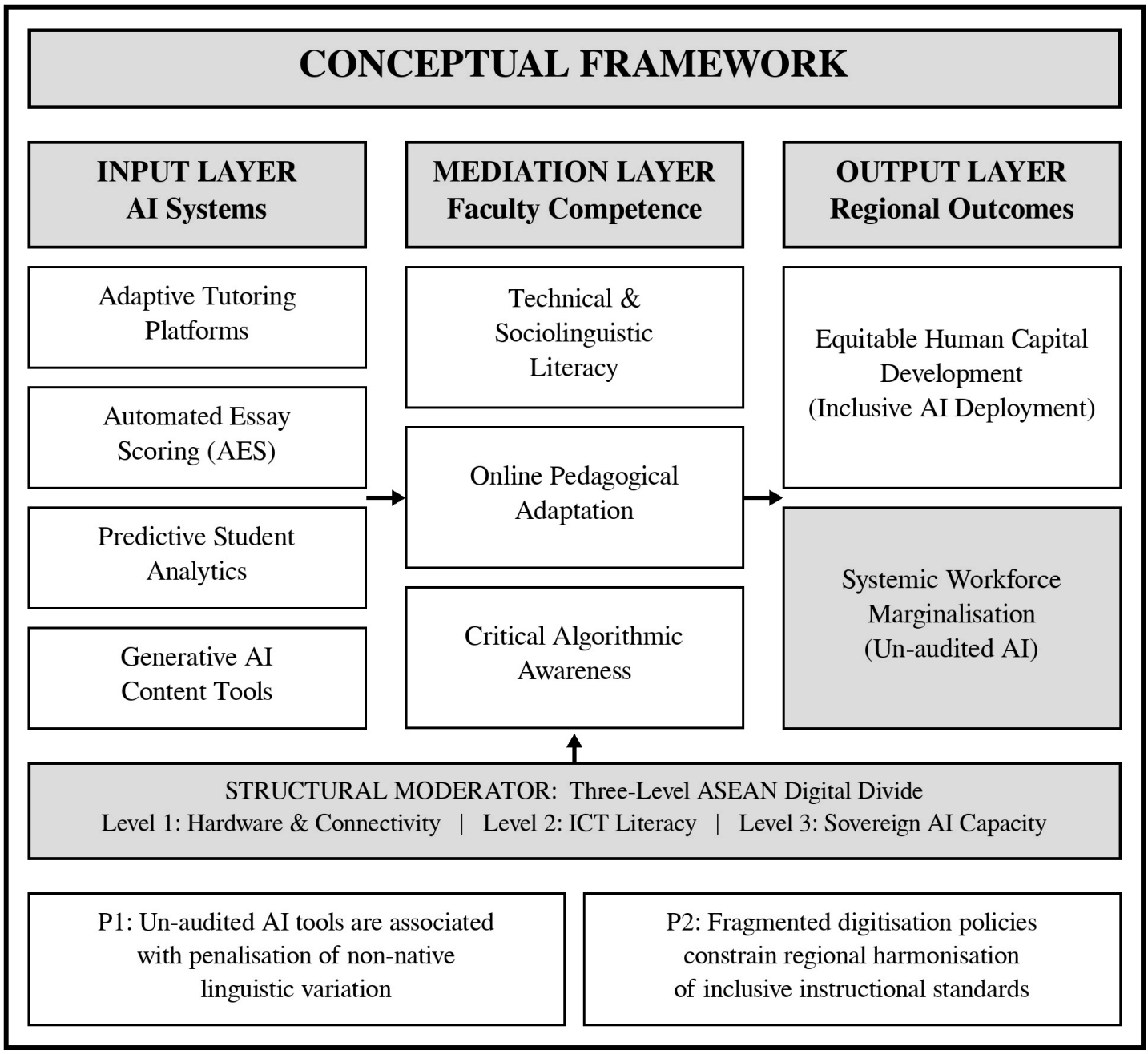


Figure 1

Conceptual Framework: Three-Layer Model of AI-Faculty Mediation and Regional Educational Equity Outcomes in ASEAN

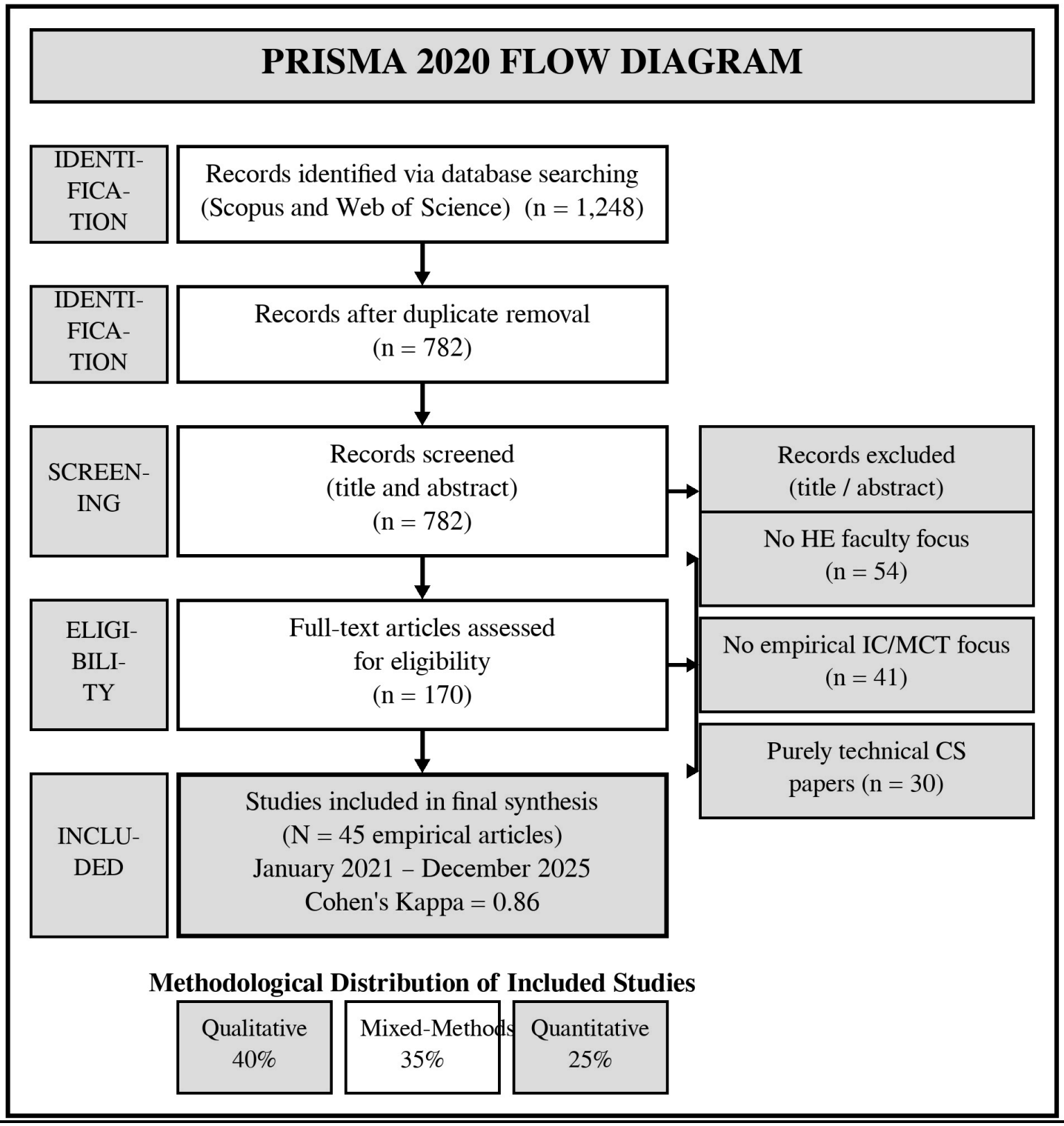


Figure 2

PRISMA 2020 Flow Diagram: Systematic Search, Screening, and Inclusion Process (N = 45)